FOURIER SERIES & FOURIER TRANSFORM | Unified Quick Reference

Engineering Mathematics • Discrete Harmonics ↔ Continuous Spectrum • Visual Enhancement Edition

1 RELATIONSHIP: SERIES ↔ TRANSFORM

Fourier Series (FS) expands a *periodic* function (or a function on a finite interval) as a discrete sum of harmonics. **Fourier Transform (FT)** represents an *aperiodic* (or transient) function as a continuous superposition of complex exponentials.

Limiting link: if a periodic function of period $2L$ has coefficients c_n, then as $L \to \infty$ the discrete spectrum c_n becomes the continuous density $F(\omega)$. FS is the tool for finite domains / periodic signals; FT is the tool for the whole line / aperiodic signals.

2 FOURIER SERIES (PERIOD $2L$)

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right)$$

$$a_0 = \frac{1}{L}\int_{-L}^{L} f\,dx, \quad a_n = \frac{1}{L}\int_{-L}^{L} f\cos\frac{n\pi x}{L}\,dx, \quad b_n = \frac{1}{L}\int_{-L}^{L} f\sin\frac{n\pi x}{L}\,dx.$$

Complex form: $f(x) = \sum c_n e^{in\pi x/L}, \quad c_n = \frac{1}{2L}\int_{-L}^{L} f\,e^{-in\pi x/L}\,dx.$

Half-range (on $(0, L)$):

Odd extension → sine series: $b_n = \frac{2}{L}\int_0^L f\sin\frac{n\pi x}{L}\,dx$

Even extension → cosine series: $a_n = \frac{2}{L}\int_0^L f\cos\frac{n\pi x}{L}\,dx$

3 FS CONVERGENCE & GIBBS

Piecewise smooth f: converges to $f(x)$ at continuity points; to the average of left/right limits at jumps. Near a jump the partial sums overshoot by $\approx 9\%$ of the jump (Gibbs). Term-by-term differentiation requires extra regularity.

Gibbs note. Near a jump discontinuity, Fourier partial sums exhibit overshoot. Increasing the number of terms narrows the oscillatory region, but the limiting overshoot does not vanish.

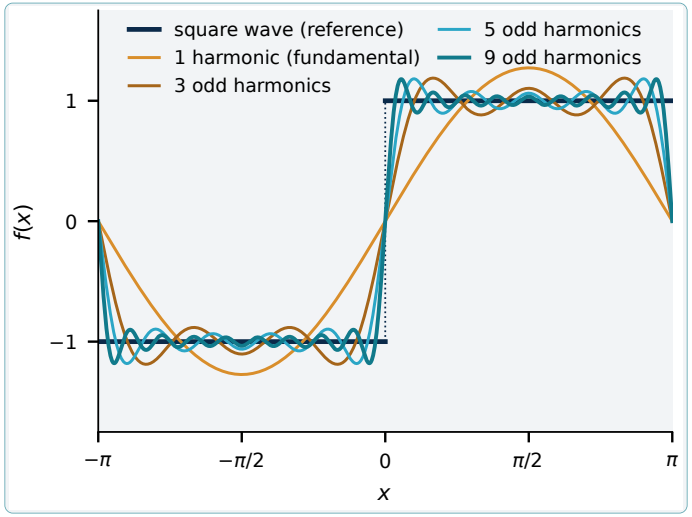
Existence (FS): piecewise smooth on one period is sufficient.

4 SYMMETRY & ENERGY (FS)

FS Parseval: $\frac{1}{L}\int_{-L}^{L}|f|^2 = \frac{|a_0|^2}{2} + \sum\left(|a_n|^2 + |b_n|^2\right).$

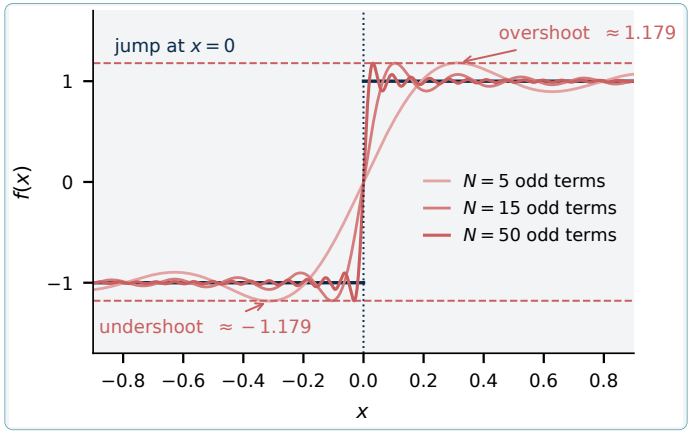
A HARMONIC RECONSTRUCTION OF A SQUARE WAVE

$f(x) = \frac{4}{\pi}\left[\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \cdots\right]$; $f = +1$ on $(0, \pi)$, -1 on $(-\pi, 0)$



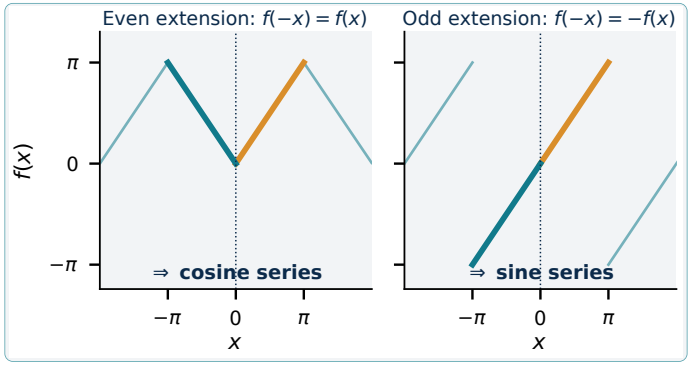
Partial sums $S_N(x) = \frac{4}{\pi}\sum_{k=1}^{N}\frac{\sin(2k-1)x}{2k-1}$ with $N = 1, 3, 5, 9$ odd harmonics. More harmonics follow the reference square wave more closely away from the jumps.

B GIBBS PHENOMENON



Zoom at the jump $x = 0$. The maximum of S_N occurs near $x = \pi/(2N)$ and tends to $\frac{2}{\pi}\mathrm{Si}(\pi) \approx 1.179$, i.e. about 9% of the jump 2. More terms squeeze the ringing toward the jump but do not lower the peak.

C EVEN AND ODD EXTENSIONS



Given $f(x) = x$ on $(0, \pi)$ (orange). **Even:** $f(-x) = f(x)$, giving $|x|$ and a *cosine* series. **Odd:** $f(-x) = -f(x)$, giving x and a *sine* series. Teal: extension and its periodic repeats.

5 QUICK PDE LINK (SEPARATION OF VARIABLES)

Homogeneous BCs on a finite interval → spatial eigenvalue problem whose eigenfunctions are sines or cosines.

- Dirichlet $u(0) = u(L) = 0$ → **sine** series
- Neumann $u_x(0) = u_x(L) = 0$ → **cosine** series

Expand initial (or remaining boundary) data in that eigenbasis; coefficients become the coefficients of the time-dependent (or second spatial) factors. Higher modes decay faster in the heat equation; they oscillate at harmonic frequencies in the wave equation.

6 METHOD SELECTION GUIDE

Situation	Use
Periodic signal / finite interval	Fourier Series
Aperiodic / whole-line / transient	Fourier Transform
PDE on finite domain (heat, wave, Laplace)	Separation of variables + FS
LTI system, filtering, modulation	FT + convolution theorem
Even/odd data on $(0, L)$	Half-range cosine / sine series
Impulse or constant signals	Generalized FT (distributions)

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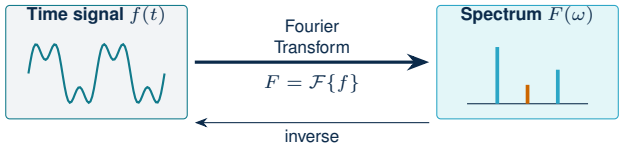
7 FOURIER TRANSFORM (ENGINEERING CONVENTION)

Forward: $F(\omega) = \int_{-\infty}^{\infty} f(t)\, e^{-i\omega t}\, dt$

Inverse: $f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)\, e^{i\omega t}\, d\omega$

$\omega = 2\pi f$ (rad/s). Other conventions exist; this sheet is consistent throughout.

D TIME DOMAIN ↔ FREQUENCY DOMAIN



$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t}\, dt$ expresses the signal in terms of its **frequency components**: $|F(\omega)|$ measures how much of frequency ω is present and $\arg F(\omega)$ gives its phase.

8 FUNDAMENTAL FT PAIRS

$f(t)$	$\leftrightarrow F(\omega)$
$\delta(t)$	$\leftrightarrow\ 1$
1	$\leftrightarrow\ 2\pi\delta(\omega)$
$e^{-at}u(t),\ a>0$	$\leftrightarrow\ 1/(a+i\omega)$
$e^{-a\lvert t\rvert},\ a>0$	$\leftrightarrow\ 2a/(a^2+\omega^2)$
$\cos\omega_0 t$	$\leftrightarrow\ \pi[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$
$\sin\omega_0 t$	$\leftrightarrow\ (\pi/i)[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]$
$\mathrm{rect}(t/T)$	$\leftrightarrow\ T\,\mathrm{sinc}(\omega T/2\pi)$

$\mathrm{sinc}(x) = \sin(\pi x)/(\pi x)$, $\mathrm{sinc}(0) = 1$. Pulse width $T \Rightarrow F(\omega) = 2\sin(\omega T/2)/\omega$.

9 SYMMETRY & ENERGY (FT)

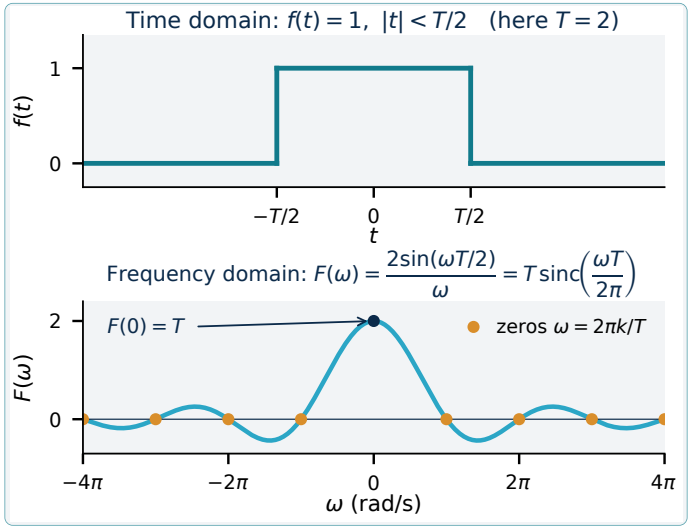
FT (real f): $F(-\omega) = F^*(\omega)$ (Hermitian). Even $f \Rightarrow F$ real & even; odd $f \Rightarrow F$ pure imaginary & odd.

Plancherel: $\int |f(t)|^2 dt = \frac{1}{2\pi} \int |F(\omega)|^2 d\omega.$

10 FT PROPERTIES

Property	Time	Transform
Linearity	$af + bg$	$aF + bG$
Time shift	$f(t - t_0)$	$e^{-i\omega t_0} F$
Freq. shift	$e^{i\omega_0 t} f$	$F(\omega - \omega_0)$
Scaling	$f(at)$	$\frac{1}{\lvert a\rvert} F(\omega/a)$
Time rev.	$f(-t)$	$F(-\omega)$
Diff. in t	f'	$i\omega F$
Mult. by t	$tf(t)$	$i\, dF/d\omega$
Convolution	$f * g$	FG
Product	fg	$\frac{1}{2\pi}(F * G)$

E RECTANGULAR PULSE AND SINC SPECTRUM

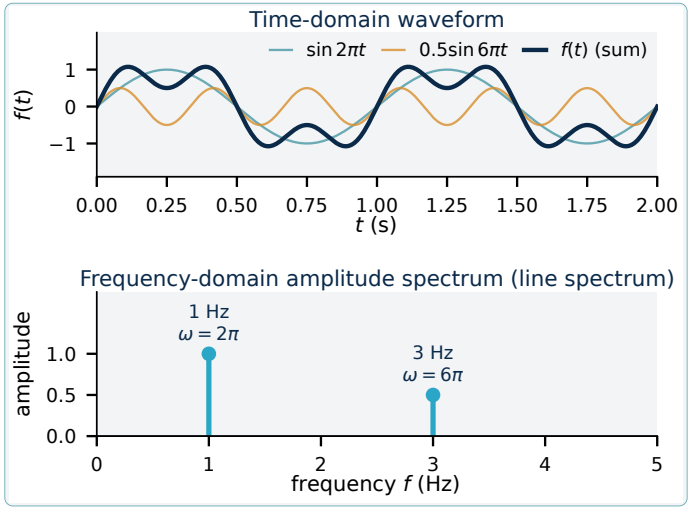


With $F(\omega) = \int f e^{-i\omega t} dt$: $F(0) = T$ equals the area of the pulse. Zeros at $\omega = 2\pi k/T$ ($k \neq 0$); a narrower pulse gives a wider spectrum.

11 EXISTENCE CONDITIONS (FT)

FT (classical): absolute integrability $\int |f| < \infty$ is the standard sufficient condition. Local BV + decay at infinity also works. Many signals (sinusoids, step) exist only as tempered distributions; δ-pairs are understood in that sense.

F FREQUENCY SPECTRUM OF A COMPOSITE SIGNAL



$f(t) = \sin(2\pi t) + 0.5\sin(6\pi t)$: two sinusoids give two spectral lines, at 1 Hz (amplitude 1) and 3 Hz (amplitude 0.5). Computed by FFT over a whole number of periods.

12 ENGINEERING APPLICATIONS

- **LTI systems and filtering:** output = input $*$ impulse response, so in frequency $Y = HX$ (convolution theorem).
- **Modulation:** multiplying by $e^{i\omega_0 t}$ shifts the spectrum by ω_0 (frequency shift).
- **Signal analysis:** spectrum lines identify frequency content; pulses give sinc spectra.
- **Finite-domain PDEs:** FS via separation of variables.

13 COMMON ERRORS: QUICK CHECK

- Using FS for a genuinely aperiodic signal (or FT on a finite interval without extension).
- Wrong half-range choice (sine for Neumann, cosine for Dirichlet).
- Omitting $1/2\pi$ in the inverse FT or the factor $2/L$ in half-range coefficients.
- Sign errors in shift theorems; forgetting $|a|$ in the scaling property.
- Treating δ-function pairs as ordinary functions.
- Differentiating an FS term-by-term without justification.
- Confusing ω (rad/s) with ordinary frequency f (Hz).
- Evaluating $2\sin(\omega T/2)/\omega$ by direct substitution at $\omega = 0$.

Convention (FT): $F(\omega) = \int f(t) e^{-i\omega t} dt$, $f(t) = \frac{1}{2\pi} \int F(\omega) e^{i\omega t} d\omega$. FS and FT are consistent spectral tools; choose according to the domain and periodicity of the problem.